

1.	$\mathbb{F} 1, ((A \wedge \neg \Box A) \rightarrow \Diamond \Box (A \wedge \neg \Box A)) \rightarrow (A \rightarrow \Box A) \checkmark$	Assumption
2.	$\mathbb{T} 1, (A \wedge \neg \Box A) \rightarrow \Diamond \Box (A \wedge \neg \Box A) \checkmark$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, A \rightarrow \Box A \checkmark$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, A$	$\rightarrow \mathbb{F} 3$
5.	$\mathbb{F} 1, \Box A \checkmark$	$\rightarrow \mathbb{F} 3$
6.	$\mathbb{F} 1.1, A$	$\Box \mathbb{F} 5$
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7.	$\mathbb{F} 1, A \wedge \neg \Box A \checkmark$ $\mathbb{T} 1, \Diamond \Box (A \wedge \neg \Box A) \checkmark$	$\rightarrow \mathbb{T} 2$
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8.	$\mathbb{F} 1, A$	$\wedge \mathbb{F} 7$
9.	$\mathbb{F} 1, \neg \Box A \checkmark$	$\neg \mathbb{F} 8$
10.	$\mathbb{T} 1, \Box A$	$\Diamond \mathbb{T} 7$
11.	$\mathbb{T} 1.2, \Box (A \wedge \neg \Box A)$	$\mathbb{T} \Box 10$
12.	$\mathbb{T} 1.2, A \wedge \neg \Box A \checkmark$	$\wedge \mathbb{T} 11$
13.	$\mathbb{T} 1.2, A$	$\wedge \mathbb{T} 11$
14.	$\mathbb{T} 1.2, \neg \Box A \checkmark$	$\neg \mathbb{T} 13$
15.	$\mathbb{F} 1.2, \Box A \checkmark$	$\Box \mathbb{F} 14$
16.	$\mathbb{F} 1.2.1, A$	$\Box \mathbb{T} 10$
17.	$\mathbb{T} 1.2.1, A \wedge \neg \Box A \checkmark$	$\wedge \mathbb{T} 16$
	$\mathbb{T} 1.2.1, A$	
	$\otimes$	

1.	$\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond (A \vee B) \checkmark$	Assumption
2.	$\mathbb{T} 1, \Diamond A \vee \Diamond B \checkmark$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond (A \vee B)$	$\rightarrow \mathbb{F} 1$
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4.	$\mathbb{T} 1, \Diamond A \checkmark$ $\mathbb{T} 1, \Diamond B \checkmark$	$\vee \mathbb{T} 2$
5.	$\mathbb{T} 1.1, A$	$\Diamond \mathbb{T} 4$
6.	$\mathbb{F} 1.1, A \vee B \checkmark$	$\Diamond \mathbb{F} 3$
7.	$\mathbb{F} 1.1, A$	$\vee \mathbb{F} 6$
8.	$\otimes$	$\Diamond \mathbb{T} 8$
9.	$\mathbb{T} 1.2, B$	$\Diamond \mathbb{F} 3$
10.	$\mathbb{F} 1.2, A \vee B \checkmark$	$\vee \mathbb{F} 10$
11.	$\mathbb{F} 1.2, A$	$\vee \mathbb{F} 10$
	$\mathbb{F} 1.2, B$	
	$\otimes$	